AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I077*

Team Name: *Aether*

Domain: *4*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

How can we make E-Waste recycling in India as easy and rewarding, while ensuring it reaches authorized recyclers safely?

Problem Understanding

- Primary users / target group: People who have old or unused phones, laptops, tablets, chargers, and other electronic devices and want an easy way to dispose of or recycle them.
- Root Cause:The collection network is fragmented,valuation is mostly manual,informal channels dominate and there are few simple digital platform connecting consumers with recyclers
- Secondary stakeholders: Authorized e-waste recyclers and collection partners, device manufacturers and producer responsibility organizations under EPR rules, government and pollution control regulators, and informal collectors who could be brought into the formal system as pickup partners.

Existing Gaps & Challenges

- Unsustainable rewards: Not every user will want the same type of reward.While a college student may prefer a groupon or credit points, a household would prefer direct cash.
- Data privacy of previously used devices could be at risk making the consumers distrusting of the initiative.
- Main challenge-
  Expensive collection logistics: Door to door collection is non feasible in less densely populated areas

Our Proposed Solution

- Core Idea: Proposed digital platform that connects people who have unwanted electronic devices with collection partners, device testing shops, refurbishers, and e-waste recyclers.
- Partnering up with local collection networks and local tech shops for quality check will make the initiative more feasible and applicable in towns and sub urban areas. A highly personalized and transparent application will also give the consumer a trustworthy platform for E-waste disposal
- A rewards system that ensures credible reward for responsible E-waste disposal, while providing the customer with a transparent and trustworthy interface for refurbishing/recycling the waste in order to prevent harm to the climate.

Technical Approach

Consumer–>ReTech Platform–>Device Assessment–>Smart Routing–>Resale or Refurbishment(Reusable)/Authorized recycle(end of life devices)

Tech Stack

Frontend: HTML, CSS, JavaScript (web interface prototype)

Backend: Python (Flask) — for device registration, valuation rules, and workflow management

Database: Google Sheets for the initial prototype; SQLite for a later working application

AI/ML, APIs, or Hardware: Google Maps API for pickup routes; rules-based device valuation initially, with AI as a possible future enhancement

Innovation & Uniqueness

- Whats new is that we allow batch wise collection and transparent tracking system along with personalized reward system with an easy interface for the user, while also protecting their device privacy

- simply making a collection tool will not be reliable, which is why making a trusted and transparent system is key for the initiative to be a success.Users may have different reward preferences, might have data privacy concerns and also have doubts as to weather the waste is actually refurbished or recycled which is where our system comes in to rectify these problems.

- The key differentiator is the AI machine learning based personalized reward system and the batch wise collection for better feasible logistics along with a device tracker and package passport for safe and reliable recycling of the e-waste

Feasibility & Scalability

- **Feasibility:** The pilot budget is required for initial reward system and basic logistics, the main source of income will be from neighborhood, hostels and educational institutes which pay our entity for e wate collection and trusted deposition of it. The other source of income will be form the refurbishers/retailers
- **Scalability:** Start with hostels and small appartment communities and slowly increase local partners to expand the business into bigger towns and cities.
- **Risks:** Expensive logistics is one of the primary concerns, and consumers not following the handover guidebook which could make their data privacy at risk.

Cost for Product (Optional)

- Development or hardware cost: 2,000–5,000 (website prototype, design, and basic testing; no dedicated hardware required).
- Hosting or maintenance cost: 0–1,000 per month (free-tier hosting and basic maintenance).
- Estimated total: 2,000–6,000 for the initial prototype and first month of basic hosting/maintenance.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Consumer Disposing of an Old Device: A consumer registers an unused smartphone on ReTech, receives an estimated value, books a pickup, and earns a reward after the device is assessed and the agreed terms are confirmed.
- Use case 2: Refurbishment and Resale: A collection partner inspects a used laptop and finds it suitable for repair. The laptop is sent to a refurbishment partner and, after repairs and quality checks, can be resold, extending its useful life.
- Use case 3: Responsible E-Waste Recycling: A consumer submits a damaged, unusable electronic device. ReTech routes it to an appropriate authorized recycler and records the handovers in its digital Recycle Passport, improving traceability and accountability.

Impacts and Benefits

- A consumer registers an unused smartphone on ReTech, receives an estimated value, books a pickup, and earns a reward after the device is assessed and the agreed terms are confirmed.
- A collection partner inspects a used laptop and finds it suitable for repair. The laptop is sent to a refurbishment partner and, after repairs and quality checks, can be resold, extending its useful life.
- A consumer submits a damaged, unusable electronic device. ReTech routes it to an appropriate authorized recycler and records the handovers in its digital Recycle Passport, improving traceability and accountability.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 12	Target 12.5	Responsible Consumption and Production	ReTech promotes e-waste reduction through device reuse, refurbishment, recycling, and responsible disposal.
SDG 11	Target 11.6	Sustainable Cities and Communities	ReTech supports cleaner cities by improving e-waste collection and directing discarded electronics to appropriate authorized recyclers.

Individual Member Contribution

Member	Contribution or responsibility
Aagney Kanjur Vinod	Assignment, problem statement and chart work
Adithya Arun	research, chart work, Business model design
Nandana Arun Nair	Team leader, Chart work, Methodology design
Pallavi S Nair	Assignment, Chart work, Fault detection

Yukti ID

IR2026-1051048